COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-DAT-001: Enterprise Data Access and Integration Services

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Title	Enterprise Data Access and Integration Services
Executive Sponsor	Chief Data Officer
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, including employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
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1. Purpose

OIT is statutorily obligated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy stakeholder needs surrounding application integration for the enterprise, this document is submitted as the standard for providing Enterprise Data Access and Integration Services.

This document communicates the standards OIT has developed with the intent of providing enterprise standard configurations for communication between heterogeneous systems within the enterprise, primarily for data exchange and service governance, based on a best-of-value decision assessment involving stakeholder needs, hardware configurations, and operational and security best practices.

The ultimate purpose of this standard is to describe the minimum requirements for designing, installing, securing, monitoring, maintaining, protecting, and decommissioning the application integration and data access services for the enterprise.

2. Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

This standard supports OIT's intentions for a unified data management strategy and applies to any technical requirements for providing application access to data, data integration, message brokers, or other means of access and interaction with solution data stores. It establishes enterprise-wide expectations for conformance to OIT's technical specifications and support for its established API-centric and Enterprise Service Bus (ESB) platforms.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

Chief Data Officer

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Authority

All data integration requirements and skills are under the authority and collaboration of the OIT Data Integration Services Team and the Chief Data Officer (CDO).

- The skills, roles, and technology required for the deployment and support of any data access, data integration, API, ESB, or application message brokers, are subject to review and approval by the OIT Data Integration Services Team and the CDO.
- Any data integration or access requirements should subscribe to an API-centric approach; development of API connectivity is the primary objective of any data access or integration requirement.

Preference and Priority of an API-centric Approach

- The State's Strategy for Data Integration and Sharing is API-led connectivity.
- The OIT MuleSoft platform shall be used for the Integrated Development Environment and full API lifecycle management.
- Support for RESTful API frameworks shall be the primary architectural pattern (no SOAP); the RAML specification shall be used to provide an API definition.
- All Enterprise API Endpoints shall be managed by and integrated into OIT's MuleSoft Anypoint Management Platform, which governs API access through established OIT Enterprise Identity and Access Management processes.

- All Enterprise APIs will be published to the Colorado MuleSoft Anypoint Exchange to make API resources discoverable and reusable.
- OIT standards for metadata and governance of APIs will be adhered to.
- All API developers (vendors, contractors, employees) must review existing API assets in the Colorado MuleSoft Anypoint Exchange before developing new APIs.
- API design will be reviewed by OIT's Enterprise Integrations personnel before development begins.
- APIs will follow OIT's secure system development practices and enterprise CI/CD processes and platforms.

Generalized Properties of an API, Enterprise Service Bus, or Message Broker

Services relating to the use of an API, ESB, or Message Broker are referred to generally as the application or data Integration Layer, which must provide:

- Adaptation: Configuration of service requestors and providers within an environment of change, including where a service varies from its definition, semantically or non-semantically.
- Auditing: The ability to audit messaging between service requesters and providers, including translation, transformation, or orchestration, and the addresses of the communicators.
- Authorization / Identity Management: Authorization and/or access control independent of either requestor or provider.
- Compensation: The ability to adapt to error conditions, maintaining integrity and providing for unaffected services.
- Composition: The ability to mediate a set of messaging requests to smaller services that semantically provide for the scope of the request in aggregate.
- Logging: The ability to log interactions at the message level.
- Protocol Switching: Facilitating messaging between providers and requesters using different protocols and interaction patterns.
- Rules: Allowing for both mediation flows and integrations that are rules-based.
- Transformation: Altering messages semantically so an intended rule-driven or other interaction layer-defined configuration can be fulfilled.
- Translation: Altering messages syntactically, through mediation flows and message models, so the requester's message can be processed by the provider for the expected result.

Scope of the Integration Layer

The Integration Layer should provide for Enterprise Application Interaction, with additional features directed specifically to the properties above. Providing integrated data analytics not meant to inform components for improved performance is considered over-reaching and anti-competitive. The Integration Layer provider should allow customizations for mediation built at a “freemium”, trial, or open-source tier to be used within/migrated to the commercial context with full support.

Multi-tenancy and Cloud

- The Integration Layer should allow for isolation as “tenants” within the Integration Layer context, including applicable role-based access control (RBAC) and other physical or logical security controls and resource utilization.

- The Integration Layer should be available as an on-premise, cloud (IaaS/SaaS), and hybrid offering, and as compliance controls change, the state should enjoy the benefits expected of the State of Colorado's Cloud-first policy.
- The Integration Layer should support a “scale-out” deployment model, whereby incremental and predictable additions to supporting infrastructure and platforms provide corresponding improvements in functional resources and features.
- The Integration Layer should provide a consolidated interface for management and administration of tenants, roles, functions, components, and features.

The generalized logical view of an ESB shows two services exchanging messages through a message routing protocol, with the ESB providing mediation (semantic and syntactic handling) and internal processing for integration, rules, and transformation/orchestration — while remaining uninvolved with the business logic (service semantics) by design.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the “Owner of Standard” listed under Roles and Responsibilities.

Requests may be acquired by completing the “EXCEPTION REQUEST to Technical Standard” form and including the completed form (with all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

6. References

- OIT documents and policies - State of Colorado Style Library; TS-OEA-001: Technology Accessibility for Persons with Disabilities; Enforcement Category Chart
- State documents and policies - Colorado Brand Guidelines; C.R.S. 24-37.5-101, et seq.; IAW Technology Standard
- Third-party resources - My Oracle Support; Bernard Manouvrier and Laurent Menard (2008), Application Integration: EAI B2B BPM and SOA, Wiley-IEEE Press; “The Enterprise Service Bus, re-examined” (IBM developerWorks); Laura González and Raúl Ruggia (2010), “Towards dynamic adaptation within an ESB-based service infrastructure layer,” MONA '10, ACM.

7. Definitions

- Communication Protocols: The application of interaction patterns within an ESB, such as synchronous, unidirectional, and publish/subscribe.
- Cross-cutting Concern: A concern that cannot be decomposed from a system, resulting in scattering (duplication) and/or tangling (interdependency).
- Enterprise Application Integration (EAI): An architectural concept and component (or set of components) that deals with non-semantic and semantic mismatches between applications.
- Enterprise Service Bus (ESB): A middleware or integration component that fulfills, at minimum, the principles of Service Virtualization and Aspect-Oriented Connectivity.
- Mediation: An ESB role to intercept and process messages between the service requester and provider so the modified syntax achieves interconnectivity.
- Mediation Flows: The application of mediation patterns within an ESB to facilitate message transfer between parties; a flow is stateless, initiated only at the request of a service, and invokes only one provider.

- Message Models: The application of metamodels within an ESB regarding standards for translating content (e.g., HL7).
- Principle of Aspect-Oriented Connectivity: Agents offer cross-cutting concern mitigation (security, management, logging, auditing) on behalf of requesters and providers with little or no impact to the agents themselves.
- Principle of Service Virtualization: Service requestors and providers interacting with an ESB should not be restricted to the same protocol; transformation should reconcile differences; identities are known only to the ESB.
- Service: The implementation of the semantics associated with achieving business goals as an abstraction; a specification describes the task while realization is the actual implementation.
- Service Governance: Controls applied to what business functions are made reusable, subject to maturation and evolution, and how they are exposed.
- Service Operational Aspects: Operational metrics such as response time, throughput, and availability, reportable as quality of service.
- Service Semantics: The business task performed and a definition of the interface in terms of task-relevant business entities.
- Service Syntax: Connection mechanisms, specifically protocol, message formats, and security.